Week 1 Notes

In IEEE 802.11a the standard is to use an IFFT of size 64 with bandwidth of 20MHz,

Only 52 of these 64 are used, with the rest being unused. This is to avoid interference as 11 of the 12 unused sub-carriers are used in the edges of the spectrum (6 in the beginning and 5 in the end), the one left out is the DC carrier which is at the center of the spectrum, information is not carried here because analog circuitry can introduce unwanted dc components, and it could corrupt/interfere the data carried in this frequency.

4 of the 52 used sub-carriers, are pilot frequencies, these carry predefined symbols to help the receiver track phase variations and fix phase errors.

The sub carrier spacing (frequency space between 2 sub-carriers) is given by the following ex pression, with Nfft being = 64 and fs the channel bandwidth which is 20MHz.

$$\Delta f = \frac{fs}{N_{fft}} = 312,15kHz$$

In IEEE 802.11a, each OFDM symbol is composed of two parts: the useful symbol and the cyclic prefix (CP). The useful symbol is generated using a 64-point IFFT, resulting in 64 samples. Given a sampling frequency of 20 MHz, its duration is:

$$T_{\text{useful}} = \frac{N_{\text{FFT}}}{f_s} = \frac{64}{20 \times 10^6} = 3.2\,\mu s$$

The cyclic prefix consists of the last 16 samples of the useful symbol, which are copied and added to its beginning. Its duration is therefore:

$$T_{\text{CP}} = \frac{16}{20 \times 10^6} = 0.8\,\mu s \rightarrow T_{\text{symbol}} = T_{\text{useful}} + T_{\text{CP}} = 3.2 + 0.8 = 4\,\mu s$$

The cyclic prefix helps prevent inter-symbol interference (ISI) caused by multipath propagation. By adding a copy of the end of the symbol at its beginning, delayed versions of the signal are less likely to interfere with the useful part of the following symbol, provided the channel delay spread does not exceed the cyclic prefix duration.

Frame prefix (preamble)

The frame prefix consists of 2 main parts, the Short Training Field (STF) and Long Training Field (LTF), it comes in the beginning of the transmission.

The STF is a sequence of the same 16 samples repeated 10 times, therefore it has the size of 160 samples. And the time length of 8us.

It does this to produce a high autocorrelation value, the objective of this part is allow the receiver to detect the beginning of a transmission.

The Long Training Field (LTF) follows the Short Training Field (STF) and consists of a 32-sample Guard Interval (GI), followed by two identical 64-sample Long Training Symbols. The guard interval helps protect the training symbols against multipath propagation, while the repeated long training sequence is used for accurate symbol timing and channel estimation.

Section 2.1

GNU radio processes everything as a continuous stream of samples, we need to tag important information so the receiver knows what it is, like the beginning of a frame, and what parameters need to be used to decode it.

These tags are the Frame-start tag, the frame-length tag and the encoding scheme.

The encoding scheme tells which modulation and coding rate were used to encode it.

Section 2.2

The objective of this section is to detect the beginning of an incoming frame. As mentioned previously, every frame starts with a preamble, which contains the Short Training Field (STF).

The receiver takes advantage of the STF's periodic structure, where a 16-sample sequence is repeated 10 times. It calculates the autocorrelation between samples separated by 16 positions. If the samples are similar, the autocorrelation value becomes high, indicating that the receiver is probably receiving the STF.

The receiver also calculates the signal power over a moving window. This is necessary because the autocorrelation value depends on the amplitude of the received signal. By dividing the autocorrelation magnitude by the calculated power, the receiver obtains a normalized correlation value, making the detection less dependent on signal amplitude.

In the GNU Radio flowgraph, this process is implemented using two parallel branches. The first branch calculates the autocorrelation using a Delay block (16 samples), a Complex Conjugate block, a Multiply block and a Decimating FIR Filter. The second branch calculates the signal power using a Complex to Mag² block followed by another Decimating FIR Filter. Their outputs are then combined using a Divide block to obtain the normalized correlation.

Finally, the receiver detects a frame when the normalized correlation exceeds a predefined threshold for three consecutive samples. The authors found that a window size of 48 samples provided good results in their experiments.

To obtain a graphic like figure 2, in gnu radio I did the following circuit

